SPE-229837-MS

Accelerating Innovation Excellence: A Structured Innovation Framework and Its Quantifiable Impact in a Large-Scale Organization

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper details a structured methodology implemented and success story within a major organization to embed a pervasive innovation culture. The scope focuses on the strategic deployment and operationalization of a centralized ideation platform supported by a robust Innovation framework, designed to systematically capture, evaluate, and foster employee-driven innovation, ultimately accelerating organisational innovation excellence.

The approach involved a multi-faceted strategy. Key elements included: establishing a dedicated digital ideation platform accessible to all employees; implementing clear governance for idea submission, review, and selection supported by Subject Matter Experts and highest level of leadership; deploying targeted training programs to cultivate innovative thinking and platform utilization. Continuous feedback mechanisms, transparent communication regarding idea status, and a system for recognizing contributions were integral to driving engagement and ensuring the systematic progression of promising concepts, which have together contributed to a sustained culture of innovation.

The systematic implementation of this structured approach led to significant and measurable improvements in innovation performance. The organization attained the highest possible status for innovation accreditation, which was further validated by an independent audit conducted by a global innovation firm. Crucially, employee engagement surged, with participation levels and idea volume showed significant increase. This surge translated directly into tangible operational and sustainability enhancements. For instance, one idea involved repurposing a metal frame, which drastically reduced operational time, increasing productivity by 52% and cutting costs. Other impactful innovations included transforming scrap material into functional structures and initiating projects utilizing AI tools for advanced environmental protection strategies and operational optimization. These diverse outcomes confirm that a well-designed ideation platform, supported by cultural enablers, powerfully catalyzes a thriving innovation ecosystem and drives demonstrable excellence.

This paper presents a validated, model framework for systematically cultivating and scaling organisational innovation. It offers practicing professionals and innovation managers a proven model demonstrating how a structured ideation platform delivers quantifiable enhancements in operational efficiency (e.g., reduced operational times, increased productivity), resource circularity (e.g., scrap

valorization), and the deployment of advanced technologies, providing clear and direct benefits beyond theoretical constructs. The novelty of this program lies in its ability to address both the creative aspects of innovation and the rigor of execution.

Introduction

While innovation is often perceived as an individual trait that shapes organisational culture, large enterprises face complex challenges that hinder its effective implementation. Organizations that successfully navigate these challenges and seize opportunities evolve into innovative, efficient, and sustainable organizations. Fostering such innovation requires deliberate strategy, cross-functional alignment, and a culture that empowers continuous improvement. The development of an innovation strategy, governance systems, frameworks, digital technologies, tools and innovation culture plays a vital role in making creativity and innovation part of a systematic process in any large and complex organization.

Innovation Strategy:

It is easy to kick start Innovation Programmes focused on a few high visibility projects. This simplistic approach may deliver short-term results and does not create a long-term capability for Innovation, ultimately missing out on the potential of creating and capturing consistent new value for the organisation.

Charting an Innovation Strategy (Pisano, G. P. (2015)) for the organization sets the right expectations and intent of the Leadership. An evolving organisational Innovation Strategy acts as a constant guide and allows course correction to keep delivering value for the organization and larger industry. It is the blueprint that guides an organization in its quest to develop new products, services, or processes and a critical component that kick-starts and steers any innovation program, especially in large and complex organizations. Additionally, a well-defined innovation strategy helps in identifying and prioritizing high-impact projects, ensuring that resources are allocated to initiatives that offer the greatest potential for value creation.

Approach: Embedding Innovation Culture in a maritime logistics and transportation organisation

The authors present below the approach to creating and successfully running an Innovation Programme based on experience and observations over the last three years, with the objective to embed a culture of Innovation in the organisation in a short period of time.

The approach presented is representative of a large maritime organisation, whereby the key business operations cater to energy maritime logistics and transportation services. This may be different from an organization, for example, involved in the manufacture of large industrial equipment where, developing proprietary technologies, rapidly prototyping and testing new engineering blueprints and securing patents may be at the center of their Innovation strategy.

This approach is more suitable for organizations, where the innovation intent (Tushman, M., Smith, W. K., & Binns, A. (2016)) is to drive continuous improvement by harnessing incremental synergies, optimizing operational efficiencies, and enhancing service reliability. The aim was to embed innovation into everyday processes—leveraging data-driven insights, automation, and cross-functional collaboration to streamline workflows, reduce costs, and improve turnaround times. While occasionally exploring moonshot ideas, bordering disruptive technologies, the key focus was on scalable, practical innovations that deliver measurable value across the fleet, warehouses, ports, and supply chain.

Thus, ADNOC Logistics & Services (ADNOC L&S) has strategically embedded innovation to future-proof its operations, with direct sponsorship from the CEO and alignment to core pillars: value creation, sustainability, and profitability. The programme is executed through dedicated integrated platforms for Idea Management and Value Creation Tracking—enabling idea capture, evaluation, and implementation across business units. Structured around six innovation areas and twelve ideation categories, the framework supports both incremental and transformative ideas, including moonshot initiatives.

Leadership plays a central role, with the Innovation Committee (comprising of senior leaders), ensuring governance and resource allocation. Innovation planning follows the Three Horizons Model (Coley, S, McKinsey & Company, 2009), addressing continuous improvement, growth, and disruptive technologies via the AI Digital and Technology Programme. Ideas undergo a rigorous multi-stage evaluation and are tracked through digital dashboards.

Support systems include an annual innovation budget, a dedicated innovation team, and tailored training. Communication is driven through digital communities, newsletters, and townhalls, and also through a Digital Rewards & Recognition Platform. Continuous improvement is embedded through feedback loops, benchmarking, and lessons learned.

The programme has matured significantly, achieving high participation and exceeding performance targets. With robust risk management and external collaborations, the organisation has cultivated a culture of innovation that is systematic, inclusive, and results driven.

Proposed Innovation Framework

A successful innovation ecosystem requires a structured framework (ISO 56002:2019 Innovation management) that aligns strategic intent with operational execution. Based on practical application and alignment with global best practices, the proposed framework integrates six essential components. As illustrated in Figure 1, each component is critical for systematically managing the cycle from idea conception to value creation.

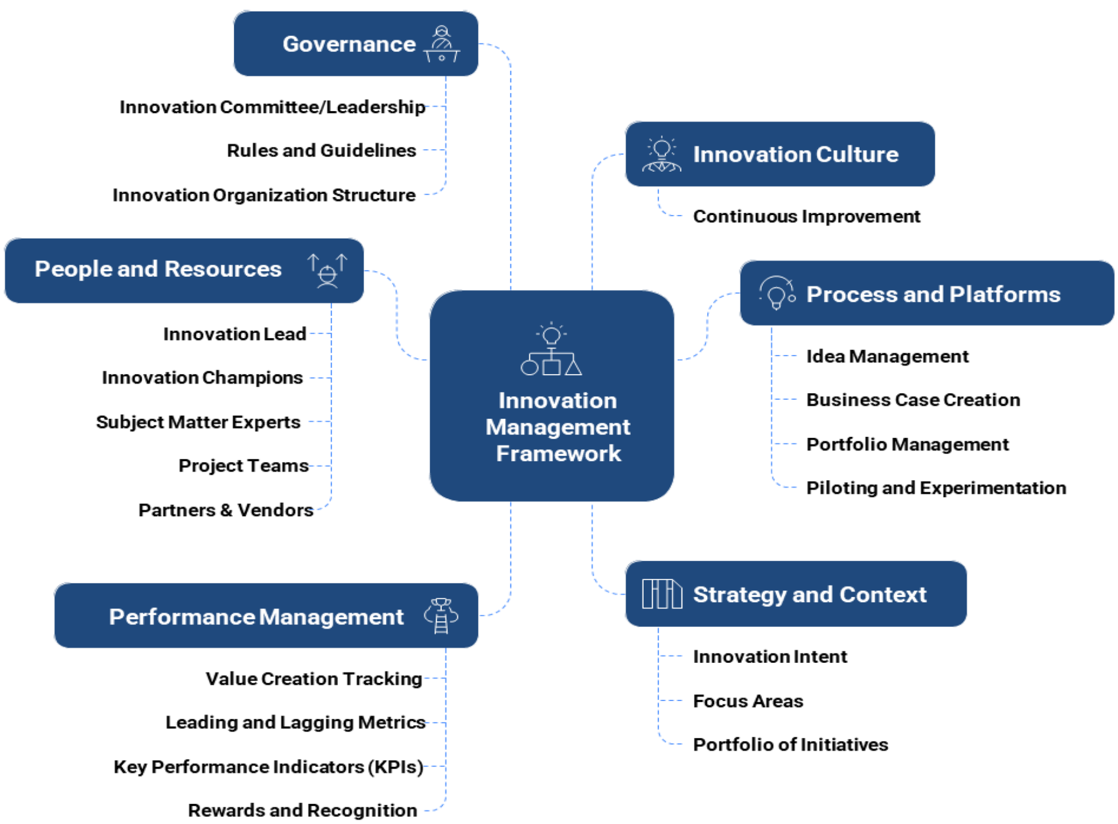

Figure 1—Proposed Innovation Framework and its components

The framework outlines the strategic architecture for building a sustainable innovation program. To illustrate its practical application, the following sections detail several key components of the framework that were deployed to accelerate progress within this structure.

1. **Strategy and Context:** The foundation of any innovation program is a clearly defined strategic intent. An organization must first articulate what it aims to achieve through innovation, identifying focus areas aligned with corporate goals such as sustainability, profitability, and operational excellence. This involves developing a balanced portfolio of initiatives that spans a range of innovation types, from incremental process improvements to transformative, long-term projects. This strategic alignment ensures that innovation is not a peripheral activity but a core driver of business value across all operational units.
2. **Governance:** Effective governance provides the structure required to steer and sustain innovation efforts. This is achieved by establishing an Innovation Committee composed of senior leaders who offer oversight, sponsorship, and strategic direction. Clear rules and guidelines must be instituted to standardize innovation practices, while a formal organisational structure should define roles and responsibilities for innovation leads, champions, and support teams. This governance model ensures accountability, transparent resource allocation, and alignment with broader business objectives.
3. **Process and Platforms:** To operationalize its strategy, an organization needs robust processes and integrated digital platforms. These systems must manage the full innovation lifecycle, from idea generation and screening to evaluation, implementation, and impact tracking. Standardized tools for business case development and portfolio management help prioritize initiatives and fast-track high-potential ideas. Furthermore, a structured approach to piloting and experimentation allows for the validation of concepts before full-scale deployment. Together, these processes and platforms ensure that innovation is systematic, scalable, and measurable.
4. **People and Resources:** Innovation is fundamentally a human-centric activity that thrives on collaboration and expertise. Organizations should formally appoint dedicated innovation leads and champions to drive engagement and build momentum. The formation of cross-functional project teams, supplemented by subject-matter experts (SMEs), brings the necessary depth and diversity to problem-solving. External partnerships with universities, suppliers, and research centers can also inject novel perspectives and specialized capabilities. Investing in both internal talent and external collaborators builds a resilient innovation ecosystem capable of delivering sustained impact.
5. **Performance Management:** To validate the success of an innovation program, organizations must define clear key performance indicators (KPIs) and track value creation. A combination of leading metrics (e.g., engagement levels, idea pipeline) and lagging metrics (e.g., cost savings, revenue generation) provides a holistic view of both progress and outcomes. Real-time performance dashboards can offer transparency into innovation activities, while structured rewards and recognition programs incentivize participation and celebrate success. This systematic approach to performance management aligns innovation efforts with strategic goals and reinforces a results-oriented culture.
6. **Culture and Continuous Improvement:** A vibrant innovation culture is characterized by psychological safety, openness to experimentation, and a commitment to continuous learning. Organizations must create an environment where employees are encouraged to provide feedback, celebrate successes, and learn from failures. Continuous improvement can be driven systematically through external benchmarking, platform enhancements, and the integration of new technologies. By embedding these principles into the organization's cultural fabric, companies can build the agility and resilience needed to remain competitive and future-ready.

The successful implementation of this framework was accelerated by several targeted initiatives designed to foster engagement and improve idea quality. The following five sections describe specific activities and techniques (from themed campaigns to a structured rewards program) that were instrumental in transforming the framework from a theoretical model into a dynamic, results-driven system.

Themed Ideation Campaigns to Drive Focused Innovation and Enrich the Innovation Portfolio. To channel creative efforts toward specific business priorities, the organization launched themed ideation campaigns. These campaigns focused on strategic categories such as Operational Efficiency; AI, Digital, and Technology; HSE & Sustainability; and People & Well-Being. This approach created a focused outlet for innovation, resulting in a doubling of idea submissions compared to the previous year.

The analysis of campaign data provided critical insights, enabling the innovation team to identify organisational "hotspots" of high engagement and map out areas of deep operational expertise. The Campaign witnessed doubling of Ideas submitted compared to previous year and saw hundreds of employees submitting novel ideas. As illustrated in Figure 2, this data-driven approach helps refine the organization's Innovation Portfolio (Nagji, B., & Tuff, G. (2012)) by revealing where creative potential is strongest. Furthermore, by categorizing submissions, the team identified recurring themes and opportunities for improvement across those themes.

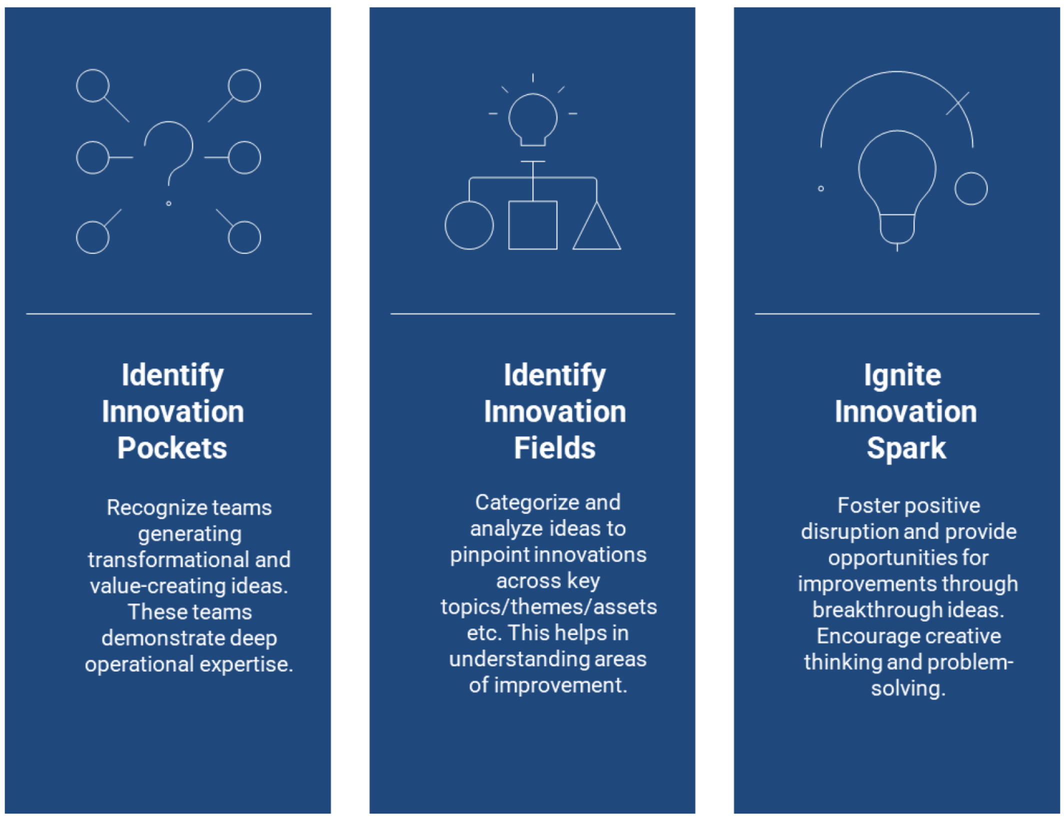

Figure 2—Role of themed idea campaigns in driving an innovative culture and refining innovation strategy

Moreover, through the collective experience and intelligence of employees, the Innovation Team identified multiple fields of Innovation. For example, categorizing and analyzing the ideas helped identify innovations and improvements across the Key Equipment types. Figure 3 below shows some key equipment types and possible innovation areas specific to maritime logistics themes

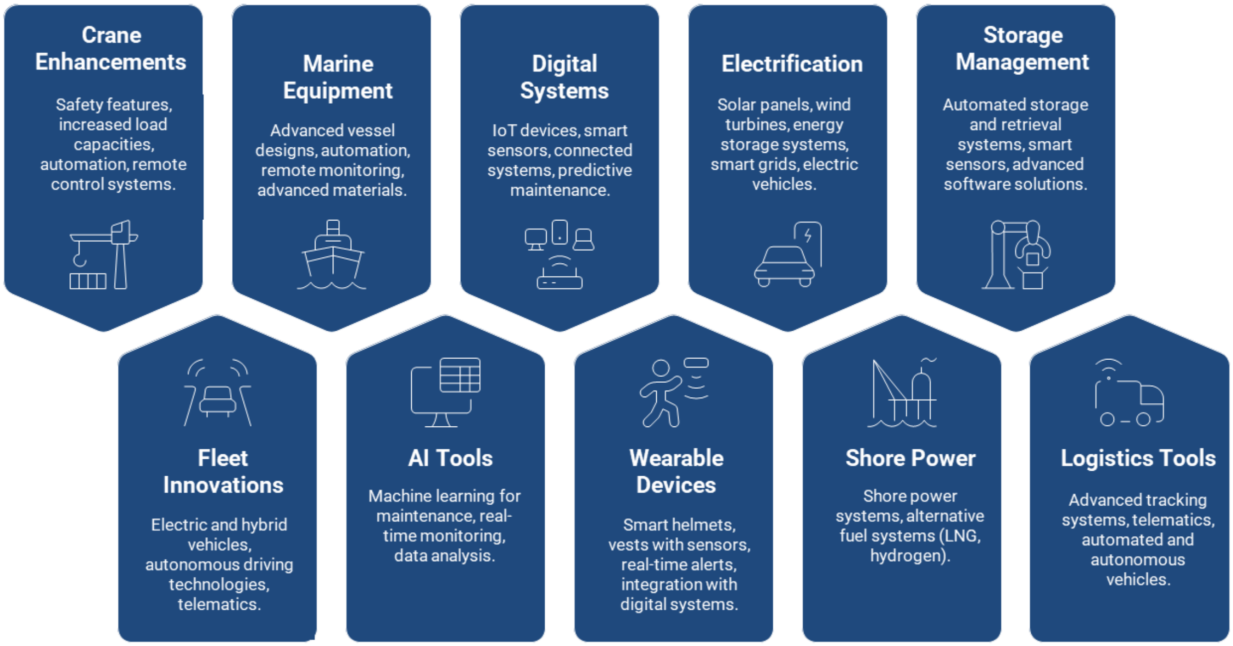

Figure 3—Example showing equipment types in a maritime logistics organisation and suggested enhancements

Catalyzing Innovation: The Role of Personalized Attention to Ideas and Idea Teams in Early-Stage Program Success. In the early stages of the Innovation Programme, personalized attention to ideas and idea teams was essential. It helped build trust, boosted morale, and encouraged active participation. By recognizing individual contributions and tailoring support to each team's context, we fostered ownership and accountability. This approach accelerated idea refinement and ensured alignment with strategic goals. Personalized engagement also revealed unique strengths, allowing us to provide targeted mentoring and allocate resources effectively. As a result, teams felt valued and motivated, which enhanced creativity and collaboration. This laid a strong foundation for a sustainable innovative culture and drove early momentum.

It is to be noted this method may not be applicable to mature Innovation programmes with a far greater number of ideas and complexity, given the fact that the Innovation Program resources may be limited. Thus, in the longer term, a robust Innovation Framework with Prioritized Portfolios, company-wide Subject Matter Experts, Innovation Champions and Input and Output KPIs make the innovation process sustainable

Structured Ideation Training. To elevate the quality and originality of employee contributions, structured training in formal ideation techniques is critical. These techniques are designed to help teams break habitual thinking patterns, challenge underlying assumptions, and explore non-obvious solutions. By providing a shared vocabulary and framework for creative problem-solving, such training fosters cross-functional collaboration and encourages risk-free exploration of unconventional concepts. While the selection of a specific technique is context-dependent, training employees in a variety of structured methods is a crucial investment in an organization's creative capability.

Figure 4 provides examples of ideation techniques and their application within a maritime logistics context.

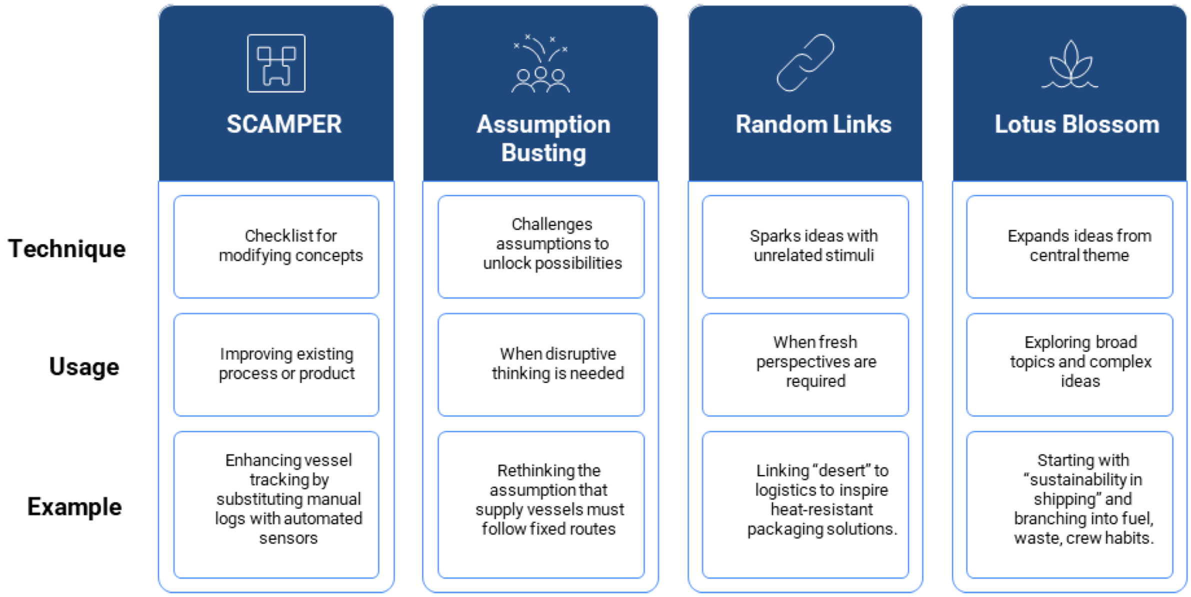

Figure 4—Ideation Techniques and examples in a maritime logistics context

Measuring and Fostering an Innovation Culture. To guide cultural development, the organization deployed an annual Innovation Culture Survey. This survey serves as a diagnostic tool to measure employee perceptions across key dimensions that enable a sustainable innovation culture. Based on the framework developed by the Dubai Government Excellence Program (Dubai Government Excellence Program. (2019)), the survey assesses aspects such as psychological safety, openness to new ideas, and leadership support. The results are used to identify areas requiring targeted intervention, thereby ensuring that efforts to build a supportive culture are data-driven and focused. An example of the survey design is shown in Table 1.

Table 1—An Innovation Culture Survey Format

Aspect	Description	Rating (1–5)
Psychological Safety	Team members feel safe to express ideas without fear of judgment or failure.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Openness to Ideas	Diverse thinking is encouraged across all levels.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Collaborative Spirit	Cross-functional teamwork and co-creation are promoted.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Experimentation	New approaches are supported, and failure is seen as learning.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Leadership Support	Leaders actively champion and invest in innovation.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Continuous Learning	Curiosity, skill development, and knowledge sharing are encouraged.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
Recognition & Reward	Creative contributions are celebrated and incentivized.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Optimized Rewards and Recognition Framework. The optimized regards and recognition approach recommends offering significant rewards for top ideas to stimulate high-impact innovation, while also granting incentives for every idea submitted to maintain broad participation and engagement. Recognizing employees who consistently contribute multiple ideas—even if not all are top-ranked—ensures that effort is fairly acknowledged.

High-value ideas with detailed, feasible roadmaps should receive special recognition to emphasize strategic thinking and execution potential. Additionally, managers who actively foster innovation within their teams should be celebrated as Innovation Champions, reinforcing a culture of leadership and collaboration.

Also, business units or functions demonstrating the highest participation should be publicly recognized to encourage healthy competition and organisational alignment around innovation goals. This holistic approach as shown in Table 2 below, not only motivates individuals but also embeds innovation into the company's culture and operational fabric.

Table 2—An improvised innovation rewards and recognition scheme to sustain innovation culture

CONSIDERATIONS	REWARDS	CRITERIA
Incentivize Best Ideas	Significant Rewards for Top Ideas. Recognition by Leadership	Significant rewards catch attention, attract participation and drives healthy competition
Each Idea Counts. All Idea Submissions to be incentivized for recognizing efforts	Incentives for each Idea Submission	Volume of Ideas. Completes feedback loop encouraging repeat participation.
Employees who take extra efforts in submitting multiple ideas	Employees rewarded proportionally based idea submission	Volume of Ideas. Ensures efforts are rewarded in proportion to participation.
Employees who submit high value impact ideas with detailed and feasible roadmaps	Rewards and Leadership recognition for Best Idea Submitters	Quality of Ideas. High Impact Ideas
Managers who take extra efforts in encouraging teams in idea submissions	Recognize Managers as Best Innovation Champions/ Teams	Teamwork. Innovation Culture
Best Business Unit/Function	Recognize Best Business Units/Function with highest participation	Leadership Support. Innovation Culture

Methodology

Driving Innovation at ADNOC L&S: Methods, Procedures, and Evolution

1. **Strategic Alignment and Organisational Context**

   The innovation program was embedded as a strategic enabler, directly sponsored by the chief executive officer (CEO) to ensure top-level commitment. Program objectives were explicitly aligned with the organization's core strategic pillars: value creation, sustainability, and profitability. To facilitate execution, the program was integrated across all business units using two primary digital platforms: a centralized ideation system for capturing and evaluating ideas, and a value-creation system for tracking project implementation and impact. The framework was structured to support a wide range of innovations, from technical and commercial improvements to health, safety, and environment (HSE) and corporate transformation initiatives.
2. **Governance and Leadership Commitment**

   Leadership plays a pivotal role in the organisation's innovation journey. The Innovation Committee, comprising executive leaders, oversees the programme. The CEO's personal involvement includes recognizing contributors, guiding framework enhancements, and allocating budgets. Leaders are accessible and engaged through periodic reviews, monthly committee meetings, and town halls. They also support idea implementation by acting as sponsors and champions, ensuring strategic alignment and resource allocation.

3. **Planning and Portfolio Management**

Innovation planning is structured around the Three Horizons Model, which ensures a comprehensive approach to growth and development. Horizon 1 focuses on continuous improvement initiatives, which are meticulously tracked via the Efficiency Platform. This horizon emphasizes the importance of refining and enhancing existing processes to maintain operational excellence. Horizon 2 targets growth vectors by exploring adjacencies and new geographies, aiming to expand the company's footprint and capitalize on emerging opportunities. This horizon is crucial for driving medium-term growth and ensuring the company remains competitive in a dynamic market.

Horizon 3 is dedicated to breakthrough Ideas and disruptive technologies and is also supported by the AIDT (AI, Digital, and Technology) programme. This horizon is all about future-led transformation, leveraging cutting-edge technologies to create new business models and redefine industry standards. The AIDT programme plays a pivotal role in identifying and nurturing these disruptive technologies, ensuring the organisation stays at the forefront of technological advancements.

Each of these horizons is supported by dedicated enablers that facilitate their successful implementation. The Investment Committee provides the necessary financial backing and strategic oversight, ensuring that resources are allocated effectively. Strategy refresh cycles allow for regular reassessment and adjustment of plans to align with evolving market conditions and organisational goals. Dedicated channels for moonshot ideas encourage bold and ambitious thinking, fostering a culture of innovation.

Key Performance Indicators (KPIs) are defined at both corporate and departmental levels, ensuring that all initiatives are aligned with the company's strategic goals. These KPIs provide a clear framework for measuring progress and success, enabling continuous improvement and accountability across the organization. This structured approach ensures that the company's innovative efforts are both systematic and impactful, driving sustainable growth and long-term success.

4. **Operational Processes and Tools**

The innovation lifecycle is managed through two integrated digital platforms: Ideation Platform, which captures ideas via open channels and themed campaigns, and Efficiency Platform, which tracks value creation, roadmap development, and implementation. The process involves several stages, including idea submission, screening by the innovation team, evaluation by idea coaches, committee review, business case development, and implementation and impact validation. Fast-tracking mechanisms are in place for mature or critical ideas, allowing them to bypass early stages and move directly to business case development.

5. **Evaluation and Performance Monitoring**

Ideas are evaluated through a multi-stage process that includes initial screening, criteria-based scoring by coaches, committee endorsement, and business case validation. The evaluation criteria encompass strategic alignment, feasibility, risk, and impact. Real-time tracking of idea progress, engagement levels, and KPI achievements is facilitated by dashboards in both platforms. Performance is measured across multiple dimensions: employee engagement in ideation, Return on Investment (ROI) from implemented ideas, alignment with the company's sustainability goals and Annual Accreditation Process

6. **Support Systems: Resources, People, and Communication**

The support systems for the programme include annual funding for platforms, training, rewards, annual accreditations, with implementation budgets managed by business units to accommodate both small and large projects. The programme is backed by a dedicated innovation lead, over 25 catalysts and idea coaches, the technology and Digital team for technology initiatives, and Innovation

Committee members from leadership. Training is offered through awareness sessions, quarterly platform trainings, and both self-paced and instructor-led courses on the in- house learning platform.

A robust communication strategy ensures widespread awareness and engagement through various channels, including digital communities, video sharing platforms, digital displays, newsletters, townhalls, company website and messaging channels. Recognition is also integrated into a unique Digital Recognition Platform, offering financial reward points, certificates, and public acknowledgments to celebrate achievements and contributions.

7. **Knowledge Management and Continuous Improvement**

   Knowledge is preserved through the implementation of duplicate detection in platforms, the maintenance of archived idea repositories, and the utilization of best practices and lessons learned modules. These measures ensure that valuable information is retained, easily accessible, and effectively utilized for continuous improvement and innovation.

   AI tools are being increasingly utilized to improve duplicate idea detection, support idea generation and evaluation.

   Continuous improvement: The approach involves root cause analysis, feedback from employees, managers, and contractors, and benchmarking with peer companies and industry leaders. Recent enhancements include the integration of the AI Digital and Technology programme, new chat features in in the Efficiency Platform, updated platform interfaces, MoUs with universities, and innovation scouting and benchmarking studies.

8. **External Engagement and Benchmarking**

   The company actively collaborates with innovation consultants, startups and vendors, universities, industry peers, and participates in international conferences such as ADIPEC. These engagements are instrumental in fostering idea development, facilitating technology adoption, and promoting knowledge sharing. An annual Innovation Accreditation Process benchmarked to global companies, ensures the Innovation Programme is updated to global standards.

9. **Risk Management and IP Protection**

   Innovation risks are managed through the application of ALARP principles, integrated risk features in Efficiency Platform , pilot implementations, Investment Committee reviews, and awareness training. Intellectual property is governed by Company's IP policy, with all ideas and solutions considered property of the company, and the PMO team overseeing IP registration.

10. **Rewards and Recognition**

    The rewards framework is designed to encourage participation at every stage by offering rewards for idea submission, additional rewards for evaluation, business case approval, and implementation, as well as through Digital platforms and face to face recognition such as in company town halls. Over the past few years, significant rewards have been distributed for innovation contributors. The revised reward framework now includes provisions for awareness contributions and consistent idea submitters.

Results and Discussion

The systematic implementation of the structured innovation framework yielded significant and quantifiable improvements in both innovation output and organisational innovation culture. The program's success was validated through internal performance metrics and a formal external audit by a global innovation consultancy, which awarded the organization its highest possible accreditation status. This section presents the key results, discusses their implications, and analyses the evolution of the program's maturity.

Quantitative Impact and Performance Gains

The most direct measure of the program's success was the dramatic increase in employee engagement and idea throughput. Within a two-year period, the volume of ideas submitted to the ideation platform tripled, demonstrating a significant shift in employee participation. This surge in engagement translated directly into tangible value, with key performance indicators showing substantial gains:

- **Idea Implementation:** The number of successfully implemented ideas increased by 66% from 2023 to 2024, indicating improved efficiency in the evaluation and execution pipeline.
- **Idea Generation:** From 2023 to 2025, the program achieved a three-fold increase in idea generation, reflecting a sustained and growing culture of contribution.
- **Strategic Goals:** The program consistently exceeded its annual targets for both cost savings and sustainability improvements.

These metrics confirm that the framework did not merely encourage ideation but effectively converted that creative potential into measurable operational and financial benefits. For instance, one implemented idea involving the repurposing of a metal lifting frame reduced a critical operational task's duration, increasing team productivity by 52% while significantly cutting material costs. Other high-impact innovations included the transformation of industrial scrap into functional safety structures and the deployment of **artificial intelligence (AI)** tools for advanced environmental monitoring and process optimization. These examples illustrate the framework's capacity to deliver diverse, high-value outcomes across the organization.

Discussion: The Power of a Structured Ecosystem: The results demonstrate that a well-designed innovation ecosystem, supported by strong governance and cultural enablers, is a powerful catalyst for organisational excellence. The significant increase in idea submissions was not an isolated event but a direct consequence of the framework's integrated design. Themed campaigns provided focus, personalized coaching built confidence, and the multi-tiered rewards system created clear incentives, all of which worked in concert to lower barriers to participation.

Furthermore, the 66% increase in implemented ideas highlights the effectiveness of the structured evaluation process and dedicated support systems. By standardizing the path from concept to execution, the framework ensured that promising ideas were not lost in administrative ambiguity but were systematically advanced, resourced, and tracked. The success of projects—from simple process efficiencies to complex AI deployments—validates the framework's ability to support innovation at all scales, from incremental improvements (Horizon 1) to transformative initiatives (Horizon 3).

Evolution and Maturity of the Innovation Program: Over its lifecycle, the innovation program has evolved from a nascent initiative into a deeply integrated organisational capability. This maturation can be characterized by several key shifts:

- **From Awareness to Integration:** The program moved beyond simply promoting innovation to embedding it within strategic business planning, performance management, and operational execution.
- **From Manual to Digital:** The transition from ad-hoc, semi-digital processes to a robust, automated digital ecosystem with real-time analytics and dashboards was a critical step-change.
- **From Ad-Hoc to Structured:** The establishment of formal governance, clear KPIs, and standardized evaluation criteria replaced ambiguity with accountability and consistency.
- **From Internal to External Focus:** The program expanded its scope from soliciting only internal ideas to actively engaging with an external ecosystem of startups, academia, and industry peers to source new technologies and insights.

- **From Reactive to Proactive:** The introduction of forward-looking practices such as horizon scanning and dedicated channels for "moonshot" ideas has shifted the organization's posture from reacting to challenges to proactively shaping its future.

This evolution was not accidental but the result of a commitment to continuous improvement, driven by leadership support and a culture that embraces learning and adaptation. The framework's success lies not only in its initial design but in its capacity to evolve, making it a resilient and enduring model for driving innovation in a large-scale industrial enterprise.

Case Studies in Value Creation and Benchmarking

To demonstrate the robustness of the structured innovation framework, this section presents three representative case studies. Each illustrates the journey from ideation to implementation, supported by quantitative outcomes and benchmark comparisons against industry norms.

- **CASE STUDY 1: Enhancing Operational Efficiency Through Equipment Re-Design**

  Context & Challenge: A recurring lifting operation on offshore assets required a custom-fabricated steel frame. The original design was heavy, time-intensive to assemble/disassemble, and created logistical bottlenecks.

  Intervention: A frontline engineer submitted an idea via the digital ideation platform to re-engineer the frame into a modular, lightweight design with improved handling capacity. The idea was fast-tracked for prototyping under the innovation governance process.

  Measured Outcomes

  - **Cycle Time Reduction:** From 8 hours to 3.8 hours per operation (**>50% improvement**).
  - **Productivity Gain:** +52% in task throughput.
  - **Cost Impact:** Eliminated need for specialized lifting gear, leading to significant savings
  - **Safety:** Reduced manual handling steps by 40%, lowering ergonomic risk.

  Benchmark Context: Compared to typical implementation timelines observed across heavy-asset sectors, this initiative demonstrated a notably accelerated turnaround. The pace and efficiency of execution positioned it among the leading examples within peer organizations recognized for operational excellence.

- **CASE STUDY 2: Achieving Sustainability Goals via Resource Circularity**

  Context & Challenge: Damaged metal-polymer slings used for bundling steel pipes were routinely discarded, creating waste and cost burdens.

  Intervention: Through a sustainability-themed campaign, an idea proposed refurbishing slings via certified vendors. A pilot was launched, followed by full-scale adoption.

  Measured Outcomes

  - **Refurbishment Cost:** Over 60% lower than new purchase.
  - **Availability:** Reducing sourcing and potential operational delays.
  - **Waste Reduction:** Diversion potential of several tonnes of material from disposal in Year 1.

  Benchmark Context: By integrating certified refurbishment into operational workflows, the organization aligned its sustainability efforts with leading standards in circular economy implementation. The approach reflects a progressive stance on waste reduction and operational resilience, comparable to top-performing peers in the sector

- **CASE STUDY 3: AI-Based Oil Spill Detection and Response**

  Context & Challenge: Oil spill detection at sea is time-critical. Traditional methods often rely on manual/delayed post-incident observation, often resulting in late detection latency.

  Intervention: An AI-powered satellite detection system was deployed in partnership with a technology provider. The model was trained on historical imagery and integrated with weather and vessel movement data.

  Measured Outcomes

  - **Detection Latency:** Significant reduction in time as compared to manual detection (which may even take hours).
  - **Accuracy:** False-positive reduction vs manual methods.
  - **Operational Impact:** Enabled faster containment.

  Benchmark Context: Industry-standard manual detection latency remains in hours. This AI solution sets a new benchmark for proactive spill response in the region.

Benchmarking Innovation Maturity

To contextualize these results, the innovation programme was benchmarked against a global dataset (2021–2024). As highlighted in Table 3 and Figure 5, the organisation achieved a 56.3% overall improvement in innovation maturity, with standout gains in Support – Resources (+84.6%) and Support – People (+75%), reflecting significant investment in budgets, training, and role-based structures.

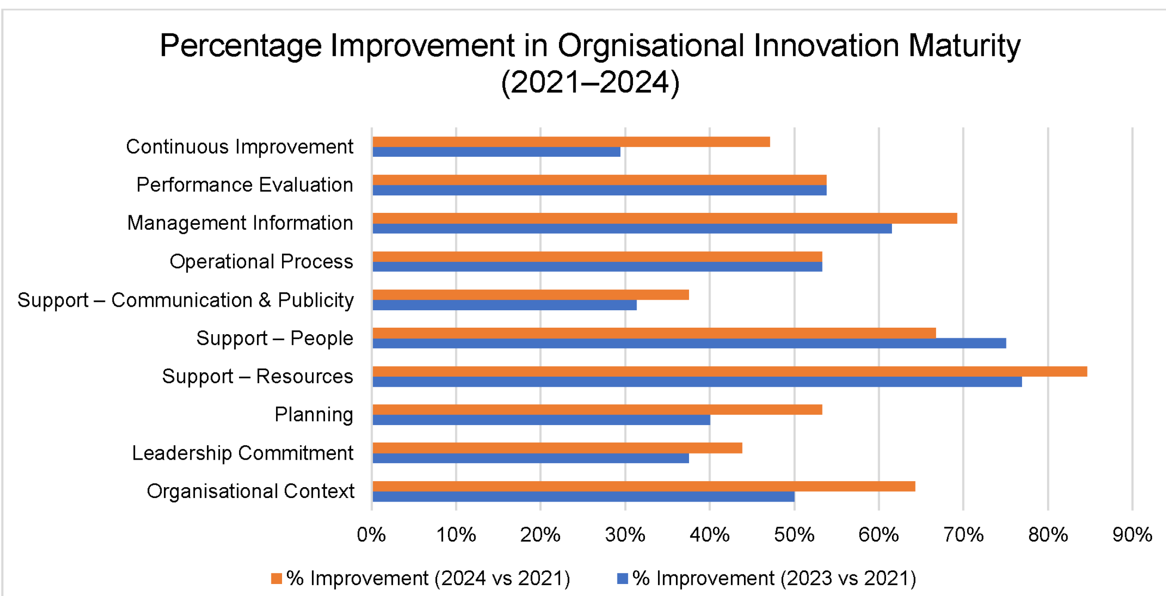

Figure 5—Integration of Innovation Framework with other organisational initiatives

Table 3—Percentage Improvement in Innovation Maturity (2021–2024) (Source: Verified Benchmarking Reports - Internal)

Dimension	% Improvement (2023 vs 2021)	% Improvement (2024 vs 2021)
Organisational Context	+50.0%	+64.3%
Leadership Commitment	+37.5%	+43.8%
Planning	+40.0%	+53.3%
Support – Resources	+76.9%	+84.6%
Support – People	+75.0%	+66.7%
Support – Communication & Publicity	+31.3%	+37.5%
Operational Process	+53.3%	+53.3%
Management Information	+61.5%	+69.2%
Performance Evaluation	+53.8%	+53.8%
Continuous Improvement	+29.4%	+47.1%
Total Score	+49.3%	+56.3%

Observation: The programme now ranks among the top quartile of accredited organisations globally, validating the scalability and maturity of its innovation framework.

In summary, these cases representing different types and scales of innovation, collectively demonstrate the framework's robustness and its ability to translate diverse ideas into tangible organisational value. Moreover, as illustrated in Figure 6, these examples above demonstrate the close Integration of the discussed innovation programme with other organisational initiatives focuses on strategic areas like Operational Efficiency, Sustainability, Culture Transformation and AI, Digital and Technology Programme.

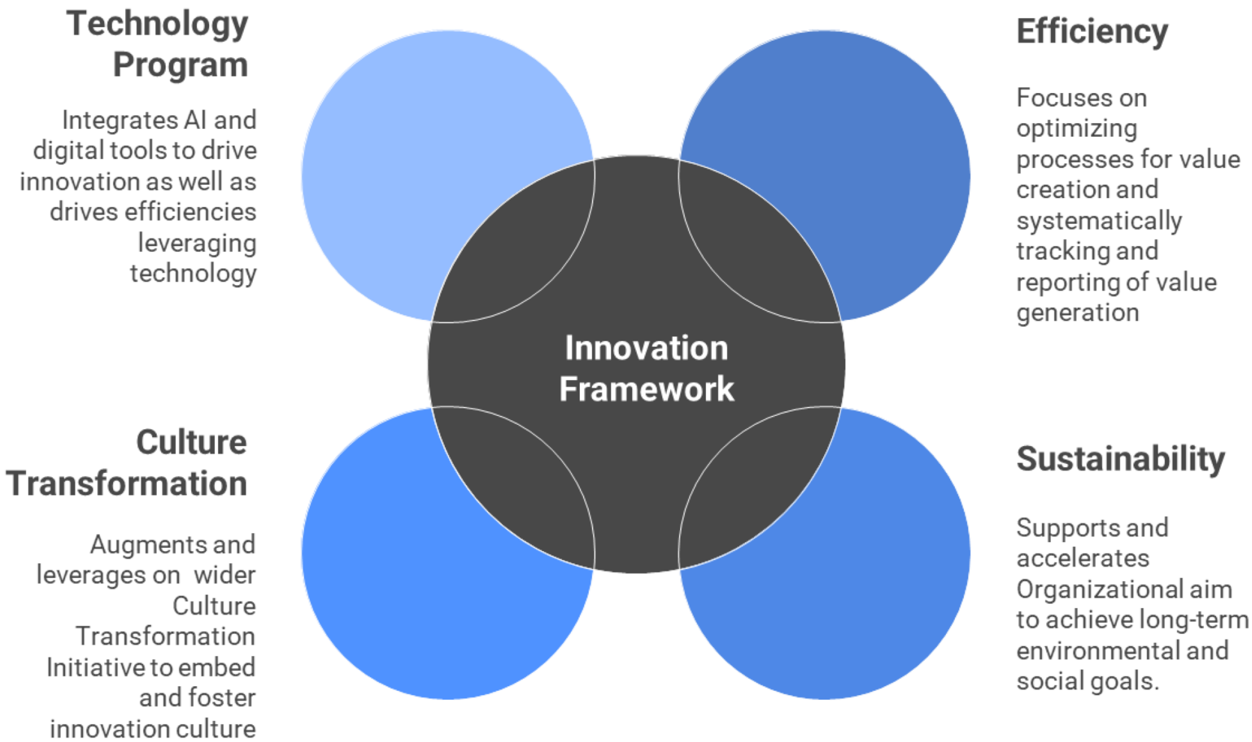

Figure 6—Integration of Innovation Framework with other organisational initiatives

Conclusion

This paper demonstrates that systematic innovation can be successfully embedded within a large-scale industrial organization through a structured, multi-component framework. The presented model's effectiveness stems from its ability to integrate top-down strategic governance with bottom-up employee engagement, creating a self-reinforcing dynamic where clear direction enables creative contribution and measurable results validate the approach.

As supported by the case studies, this integrated system is capable of translating diverse ideas into quantifiable value across critical business areas, including operational efficiency, sustainability, and technological advancement. The primary contribution of this work is the presentation of a validated, transferable model that offers a structured methodology for cultivating and scaling an internal innovation capability. It provides an empirically supported alternative to less formal or ad-hoc innovation initiatives.

The findings confirm that a deliberate and well-resourced approach which unites strategic intent, robust digital platforms, and a commitment to fostering employee potential, enables an organization to systematically convert creative concepts into tangible performance outcomes. This establishes a foundation for sustained improvement and long-term adaptability in a competitive industrial environment.

Acknowledgments

We thank all individuals who contributed their expertise and support to this study. We also acknowledge the use of AI language models in enhancing the clarity and readability of this manuscript.

References

1. Pisano, G. P. (2015, June). You need an innovation strategy. *Harvard Business Review*. https://hbr.org/2015/06/you-need-an-innovation-strategy

2. Dubai Government Excellence Program. (2019). Dubai Government Innovation Framework: Body of knowledge (Version 1.0). https://dgep.gov.ae/en/file/preview/dubai-government-innovation-framework
3. Coley, S. (n.d.). Enduring ideas:The three horizons of growth. McKinsey & Company. https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/enduring-ideas-the-three-horizons-of-growth
4. Tushman, M., Smith, W. K., & Binns, A. (2016). Domains of innovation intent: A key leadership responsibility. *Business Horizons*, **59**(6), 647–655. https://www.sciencedirect.com/sdfe/pdf/download/eid/1-s2.0-S0090261616301061/first-page-pdf
5. International Organization for Standardization. (2019). ISO 56002:2019 Innovation management — Innovation management system — Guidance. ISO. https://www.iso.org/standard/68221.html
6. Nagji, B., & Tuff, G. (2012, May). Managing your innovation portfolio. *Harvard Business Review*. https://hbr.org/2012/05/managing-your-innovation-portfolio